PRE-CLASS PREP

Gas Exchange & Oxygenation

Know Before You Go

Read this before class — it will make the lecture click faster.

Big picture: Oxygenation is foundational to every body system. Today covers altered breathing patterns, hypoxia, oxygen delivery devices, interventions to improve oxygenation, and COPD. Oxygen is a medication — it requires an order and careful titration.

ALTERED BREATHING PATTERNS

- **Hyperventilation:** ↑ RR; ↑ CO_2 exhaled → low CO_2 in blood → alkalosis; causes: anxiety, pneumonia, COPD, asthma, DKA, brain injury
- **Hypoventilation:** ↓ RR; ↑ CO_2 in blood → acidosis; causes: opioids, benzodiazepines, barbiturates, neuromuscular disorders, lung disease
- **Hypoxia:** lack of oxygen at cellular level; early signs: restlessness, ↑ HR/RR, dizziness; late: central cyanosis; interventions: supplemental O_2, monitor SpO_2

OXYGEN — KEY CONCEPTS

- **Oxygen is a medication** — requires a provider order; can be continuous, PRN, or titrated
- SpO_2 goal: **95–100%** for most patients
- **COPD patients:** different target range — hypoxic drive; use lowest effective flow
- Tanks must be secured upright; check levels regularly
- Humidification can be added to prevent dry mucous membranes

OXYGEN DELIVERY DEVICES

Nasal Cannula — 1–6 L/min; prongs in nares; monitor skin breakdown; high-flow available

Simple Face Mask — 5–10 L/min; retains some CO_2; monitor skin

Partial Rebreather — 10–15 L/min; reservoir bag; one-way valve; keep bag inflated

Nonrebreather — 10–15 L/min; highest O_2 delivery; snug fit required; **contraindicated in COPD**

Venturi Mask — 4–15 L/min; fixed precise concentration; change barrels to adjust O_2%

CPAP/BiPAP — CPAP: constant pressure (sleep apnea); BiPAP: higher inhale / lower exhale pressure (COPD, heart failure, muscle weakness)

INTERVENTIONS TO IMPROVE OXYGENATION

- **Incentive spirometer (IS):** promotes deep breathing; prevents atelectasis; 10 reps per hour, hold each breath 3–5 seconds; educate on proper use
- **Pursed-lip breathing:** breathe in through nose, out slowly through pursed lips; helps release trapped air; promotes CO_2 elimination
- **Coughing & deep breathing:** prevents atelectasis and pneumonia; use pillow to splint abdomen
- **Sputum collection:** collect before eating/drinking; 1–2 tsp in sterile cup
- **Chest physiotherapy:** percussion with vibration; loosens thick secretions; 20–30 minutes

TRACHEOSTOMY CARE

- Airway management device — requires special suctioning equipment
- Monitor SpO_2 during suctioning
- Daily care to maintain patency: cleanse stoma site and plate, change gauze under plate, replace inner cannula, change ties each day
- Open vs. closed suction systems — know facility protocol

COPD & COPD EXACERBATION

- Characterized by **persistent respiratory symptoms and airflow limitation**; pathologic changes in airways
- **Risk factors:** tobacco use, environmental/occupational exposure, respiratory infections, extreme temperatures, treatment nonadherence
- **Manifestations:** dyspnea, coughing, wheezing, fatigue, mucous changes, leg/ankle swelling, low SpO_2
- **Interventions:** O_2 if needed — **never above 6 L/min**; may require BiPAP; breathing treatments; steroids; mucolytics; pursed-lip breathing; tripoding position

DISCHARGING WITH OXYGEN

- **Walk study (6-min walk test):** RT walks with client monitoring SpO_2; determines O_2 system and flow rate needed
- Case management arranges vendor and coordinates delivery before discharge
- **Nurse educates on:** correct liter flow settings, no open flames, no smoking near O_2, keep containers secured and upright, keep away from heat, when to call provider
- Document education and return demonstration

HOME OXYGEN SAFETY

- Oxygen is combustible under certain conditions
- **Never smoke near oxygen**
- Keep away from open flames and heat sources
- Keep oxygen containers upright and secured
- Keep a fire extinguisher in the home
- No petroleum-based products (Vaseline, oil-based lotions) near oxygen equipment